

BOOK A DOCTOR SYSTEM

Configure MongoDB

This section explains how MongoDB is configured in the Book a Doctor MERN application. The backend uses MongoDB Atlas or a local MongoDB server with Mongoose for database connectivity.

Step 1: Install MongoDB Dependencies

Install Mongoose and Dotenv packages in the backend folder:

```
npm install mongoose dotenv
```

Step 2: Create .env File

Create a **.env** file in the backend root directory.

```
PORT=5000
MONGO_URI=mongodb+srv://<username>:<password>@cluster.mongodb.net/book_doctor_db
JWT_SECRET=your_secret_key
JWT_EXPIRE=7d
```

Step 3: Database Connection (src/config/db.js)

```
const mongoose = require("mongoose");

const connectDB = async () => {
  try {
    const connection = await mongoose.connect(
      process.env.MONGO_URI
    );

    console.log(
      `MongoDB Connected: ${connection.connection.host}`
    );
  } catch (error) {
    console.error(
      `MongoDB connection failed: ${error.message}`
    );
    process.exit(1);
  }
};

module.exports = connectDB;
```

Step 4: Connect Database in server.js

```
require("dotenv").config();

const app = require("./src/app");
const connectDB = require("./src/config/db");

const PORT = process.env.PORT || 5000;

const startServer = async () => {
  await connectDB();
  app.listen(PORT);
};

startServer();
```

Explanation

- Mongoose is used as the MongoDB Object Data Mapper (ODM).
- Database credentials are stored securely in the **.env** file.
- **connectDB()** establishes the MongoDB connection before the server starts.
- If the connection fails, the application exits to prevent runtime errors.
- After a successful connection, all User and Appointment models communicate with MongoDB through Mongoose.